

Distributed Knowledge



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In economic theory, companies compete and copy one another's methods, producing rough equality of efficiencies. However, firms in the same industry can differ significantly in their efficiencies and profitability in the real world. Much of strategy research is aimed at understanding this stickiness.

A company faces several management and engineering problems in adopting a “best practice” from another, like the Toyota system of manufacturing or Cirque de Soliel's ability to craft stunning shows. But, if the skills and knowledge can be codified or simply understood by a few people, they can eventually be copied by others. Fighter pilots cannot tell you all that they know about flying and fighting, but the skills can be taught, practiced, and passed on.

What cannot be moved from one organization to another or from one nation to another is what I have come to call *distributed knowledge*. Distributed knowledge exists in a group of people, each contributing a small piece of the whole. The essential economic element of distributed knowledge is that no one person is vital—there is significant redundancy, and each could be replaced by a newcomer who can soon learn their role. The essential social element is that each person knows much of their role by interacting with the people around them, taking in information, norms of behavior, and bases for judgment.

Inside the firm, distributed knowledge is often called know-how, culture, or good management. Visit several firms in an industry, and each has a distinctive character—what my one-time colleague Sumantra Ghoshal

used to call “the smell of the place.” These human systems are complicated to engineer, change, and replicate.

Among schools of business, INSEAD (France) is known for its executive development programs. During the three years I served on the INSEAD faculty, I marveled at how the school marketed, planned, and executed these programs. Each faculty member had two or three “chunks” of material they were practiced at presenting to executives and most knew about each others’ “chunks.” The marketing staff understood the intellectual elements of the programs and knew a lot about customers and potential customers. The school maintained a computerized contact file that recorded each faculty or staff contact with an executive and recorded their comments and interests. The school tracked several thousand executives in the system. Specialist program coordinators spoke several languages and took care of all the materials, logistics, travel, dining, and accommodations issues. The administration gave early program planning responsibility to young faculty members. They quickly learned that one could not efficiently recruit faculty teachers for a program unless you were also available for others’ schedules. The senior faculty and dean expected each faculty member to do some executive teaching. Unlike many U.S. schools, where such activity is restricted to a few, their attitude was clear—if you were a business school faculty member and didn’t have at least one three-hour chunk of material of interest to executives, there was something very wrong. The whole thing worked with admirable smoothness, bringing in funds for the school and exposing faculty to various executives and their viewpoints.

On returning to UCLA, I made a nuisance of myself by lecturing colleagues and administrators about how our own executive programs could benefit from INSEAD’s methods. But I made little progress because I had no realistic action plan. Each element of INSEAD’s system worked because other elements worked—there was no obvious place to crack the system open and start.

The INSEAD system was an example of *distributed knowledge*. Each faculty member, administrator, and program manager learned how to perform their role by participating in the system. Each performance depended upon the implicitly coordinated performances of others. No one knew everything, and there was no “handbook.” Knowing the basic rules and overall architecture of the system, as I did, was not enough to replicate it. It was like knowing how to drive a car but asking my colleagues to build a car from scratch. INSEAD does not need patents or secrecy to protect its skills at executive education from imitators—distributed knowledge does the job.

Distributed knowledge within the firm resists replication by other firms. Even more importantly, distributed knowledge *outside* the firm can also be a critical source of advantage. Outside the firm, distributed knowledge exists as reputation, brand image, learning, and relationships, where these apply to the ultimate buyers and suppliers, and retailers. Like employees inside the firm, people outside the firm hold knowledge and opinions about how to conduct business, choose products, and consume. And this distributed knowledge, the combined cognitive states, and the consequent behavior of hundreds, thousands, or millions of people, is a critical resource for the business. If it could be easily created through advertising or marketing efforts, others could replicate it. But a great deal of this distributed knowledge is socially complex. Nike’s advertisements have influence, but they also reach a public that has learned the unspoken rules for thinking about and consuming Nike products from each other. This distributed yet coordinated knowledge is an essential strategic asset.

Examples of distributed knowledge abound once your eyes are tuned to see it. Put on a SCUBA tank, drop fifty feet below the ocean’s surface and watch a school of small silverfish. The group is a rough sphere, and sunlight glints off each fish’s scales as they swim in one direction together. Then, like a marching squad responding to “By the left flank, turn,” each fish suddenly turns 90 degrees and darts in the new

direction, still in perfect formation. There is a lead fish in a turn who initiates the school's turn, but most of the fish do not see this lead. Instead, they watch the fish around them and respond to their movements. The turn is carried out by social coordination. And once they are moving in a new direction, a different fish assumes the role of leader. If the school members are inattentive to their neighbors, they will miss the turn and provide false signals to the other fish around them.

Other examples from the natural world are swarms of birds, bats, and bees. The video clip below shows about eighty-thousand European starlings flying in a controlled mass over the Piazza Cinquecento in Rome. Each bird takes its cue from four or five birds nearby, allowing the whole group to maintain coherence.



In human systems, coordination goes far beyond physical movements to include coordinated intelligence. The distributed knowledge and

routines underlying this coordination can be a vital source of advantage because they resist imitation, and the individuals involved are each replaceable. (Distributed knowledge vested in a group whose members could not be replaced would be equally valuable, but the value would eventually accrue to group members rather than to the firm.)

Some economists believe that organization is a matter of incentives. Give people the right rewards, they tell us, and people will perform. But exemplary organizational performance is more than the sum of individual performances; it requires flexible social coordination. The members of an organization must respond quickly to the subtle signals of their peers and continually adjust their behavior. And the knowledge of the organization—the rules about how to skillfully play each role within it—is distributed amongst its members. Here, at the mysterious heart of superior performance, we find knowledge and skill that are not knowable by anyone's mind. Call it distributed knowledge or culture, it remains a strong source of persistent advantage.